

FinalProject

Ben Thompson & Cody Coppage

2025-12-10

```
# Import Statements
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
#install.packages("fuzzyjoin")
```

```
library(fuzzyjoin)
```

```
library(lubridate)
```

```
##
```

```
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## date, intersect, setdiff, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats 1.0.0      v stringr 1.6.0
```

```
## v ggplot2 3.5.1     v tibble 3.2.1
```

```
## v purrr 1.0.4       v tidyr 1.3.1
```

```
## v readr 2.1.5
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Introduction

Extreme weather events are rapidly emerging as one of the biggest climate-driven ecological stressors worldwide (Intergovernmental Panel on Climate Change (IPCC) (2021)). Heat waves, cold snaps, ice storms, and other short-duration extreme weather events are increasing in frequency and severity, and these events can trigger massive ecological impacts in just hours or days (Intergovernmental Panel on Climate Change (IPCC) (2021); Thornton, Whitfield, and Challinor (2023)). Oregon is already experiencing rapid swings in temperature and storm intensity, with climate projections suggesting that extreme weather will become increasingly more common in the future (Oregon Climate Change Research Institute (2023)). Insects, especially bees, are extremely sensitive to quick shifts in temperature because their flight performance, metabolism, and foraging ability are directly affected by thermal limits (Heinrich (1979); Woods, Dillon, and Pincebourde (2019)). While a large amount of research shows that pollinator declines are linked to habitat loss and gradual warming trends (Potts et al. (2016)), less work has focused on what short-term extreme events do to wild bee populations. Because bees support plant reproduction and agricultural production throughout Oregon, understanding how heat waves, cold snaps, and other extreme weather events affect bee diversity is critical for conservation, ecosystem function, and resilience in the face of accelerating climate instability. Previous research has shown that extreme heat can impair thermoregulation and reduce survival in bee populations (Pereboom and Biesmeijer (2003); Woods, Dillon, and Pincebourde (2019)), while other work has shown that cold winters can alter pollinator community composition (Franzén and Öckinger (2012)). Severe winter storms have been suggested as a risk factor for bees, as cold and ice can reduce floral access and damage habitat (Mishra et al. (2023)). These challenges are happening alongside other long-term pressures like habitat loss and floral resource declines (Goulson et al. (2015)), which may make bees even more vulnerable when extreme weather occurs. However, not all bee species respond to stress in the same way. One major difference among bees is whether they are specialists or generalists. Some bee genera only use certain host plants, while others can forage from a wide range of flowers (Michener (2000); Wilson and Carril (2015); Danforth et al. (2019)). Specialists such as *Diadasia*, *Peponapis*, and *Perdita* are tightly tied to specific pollen sources (Fowler (2016); Praz, Müller, and Dorn (2018)), so if extreme weather disrupts flowering times or reduces access to those plants, specialists may take the biggest hit. Generalist bees like *Bombus*, *Lasioglossum*, and *Halictus* can use many floral resources (Wilson and Carril (2015); Goulson (2010)), which may help them to adapt to quick changes in the environment. Looking at differences between these functional groups may help explain which types are more vulnerable to extreme weather events. To address these questions in Oregon, we are using two datasets. First, the Oregon Bee Atlas dataset provides species records and sampling data across Oregon. Second, the NOAA Storm Events Database provides time and location records of extreme weather events. Because the NOAA data is categorized by county or zone and not by habitat type, our analyses will focus on Oregon-wide patterns rather than trying to isolate habitat differences. By aligning bee sampling dates with storm event windows, we can evaluate whether changes in bee diversity and functional groups occur after extreme weather events such as extreme heat, extreme cold, or ice storm events. The purpose of our study is to measure how specific extreme weather events influence bee observations across Oregon. We will compare bee data from before and after such weather events, and test whether there are differences in diversity and functional types. This analysis will help fill an information gap, because most research has focused on gradual warming trends or individual mechanisms, rather than short term weather events. The main questions we aim to answer are how does extreme heat and extreme cold events affect bee diversity across Oregon and how do these events affect different functional groups of bees. The answer to these questions will help guide conservation by identifying which types of extreme weather events are most impactful for bees and how these events affect bee composition. This could help natural resource managers target conservation efforts such as planting strategies, habitat protection, or nest site support in areas most at risk as climate extremes increase.

Dataset Identification

The data we are using includes the Oregon Bee Atlas data from 2018-2024 (<https://agsci.oregonstate.edu/bee-atlas>), and the NOAA Storm Events Database (<https://www.google.com/url?sa=t&source=>

web&rct=j&opi=89978449&url=https://www.ncei.noaa.gov/stormevents/&ved=2ahUKEwjEnvT7iK-RAxUrEzQIHUVmHtkQFnoECAQQAQ&usg=AOvVaw1IQjWMWyyZkgsoVho2eTL5). The storm events database is subset to include only extreme heat events, freezing temperatures, or winter storms in Oregon between 2018 and 2024.

The data skills we are going to use include outer joins, data cleaning with tidyverse, sampling and hypothesis testing.

The metadata (what each column of the data means) for the NOAA Storm Data is as follows: EVENT_ID: An identification number used by NOAA CZ_NAME_STR: The location of where the event occurred. Typically a zone or region, such as central Oregon or southern coast. This is the column we used for extracting the county. BEGIN_DATE: String representing the start date in format DD/MM/YYYY BEGIN_TIME: Integer data representing time of day, with values from 0 to 2400 EVENT_TYPE: Event type we queried for, includes “Heat”, “Extreme cold/wind chill”, etc... MAGNITUDE: Blank column (used for earthquakes, if we had queried for that) TOR_F_SCALE: Also blank column DEATHS_DIRECT: Integer data. 0 is used for no deaths INJURIES_DIRECT: Integer data. 0 is used for no injuries DAMAGE_PROPERTY_NUM: Integer data. Estimated number of property damage in USD DAMAGE_CROPS_NUM: Integer data. Estimated amount of crop damage in USD. STATE_ABBR: String with state abbreviation (all datas are “OR” since that’s what we queried for) CZ_TIMEZONE: All are “PST-8” MAGNITUDE_TYPE: Blank column, used for other disaster types EPISODE_ID: Unique ID used by NOAA CZ_TYPE: Location (C for county, z for forecast zone) CZ_FIPS: FIPS code for specific location WFO: Weather Forecast Office in charge of that location INJURIES_INDIRECT: Amount of indirect injuries recorded DEATHS_INDIRECT: Amount of indirect deaths recorded SOURCE: How that data was collected (public, department of highways, etc...)

```
# Importing Oregon Bee Atlas dataset
```

```
OBA_dataset <- read.csv("OBA_2018-2024.csv")
```

```
#OBA_dataset
```

```
# Importing Storm dataframe from Excel
```

```
storm_dataset <- read.csv("storm_data.csv")
```

```
#storm_dataset
```

Data Wrangling/Exploration

```
# Create "GenusSpecies" Column
```

```
OBA_dataset <- OBA_dataset %>%  
  mutate(GenusSpecies = paste(genus, specificEpithet))
```

```
#OBA_dataset
```

```
# Specifying columns to drop from storm events dataframe. These columns are not relevant for our analysis
```

```
storm_dataset_cols_to_drop <- c(  
  "EVENT_ID",  
  "CZ_NAME_STR",  
  "BEGIN_TIME",
```

```

"MAGNITUDE",
"TOR_F_SCALE",
"DEATHS_DIRECT",
"INJURIES_DIRECT",
"DAMAGE_PROPERTY_NUM",
"DAMAGE_CROPS_NUM",
"STATE_ABBR",
"CZ_TIMEZONE",
"MAGNITUDE_TYPE",
"EPISODE_ID",
"CZ_TYPE",
"CZ_FIPS",
"WFO",
"INJURIES_INDIRECT",
"DEATHS_INDIRECT",
"FLOOD_CAUSE",
"TOR_LENGTH",
"TOR_WIDTH",
"BEGIN_RANGE",
"BEGIN_AZIMUTH",
"END_RANGE",
"END_AZIMUTH",
"END_LOCATION",
"END_TIME"
)

```

```
# Dropping the useless columns from storm
```

```
storm_dataset <- storm_dataset[, !(names(storm_dataset) %in% storm_dataset_cols_to_drop)]
```

```
#storm_dataset
```

```
# Make sure the date columns are date objects
```

```
storm_dataset <- storm_dataset %>%
  mutate(BEGIN_DATE = mdy(BEGIN_DATE))
```

```
OBA_dataset <- OBA_dataset %>%
  mutate(verbatimEventDate = mdy(verbatimEventDate))
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'verbatimEventDate = mdy(verbatimEventDate)'.
## Caused by warning:
## ! 147 failed to parse.
```

```
#OBA_dataset
```

```
# Join Tables
```

```
joined_data <- fuzzy_left_join(OBA_dataset, storm_dataset,
  by = c("county" = "COUNTY",
    "verbatimEventDate" = "BEGIN_DATE"),
```

```

match_fun = list(`==`, function(bee_date, storm_date) {
  bee_date >= storm_date & bee_date <= storm_date + 30}))

#joined_data

# Count of each unique storm that occurred before a bee capture instance

joined_data %>%
  count(EVENT_TYPE)
#joined_data

# Add Control column. True if control data, false if a storm instance
joined_data <- joined_data %>%
  mutate(Control = is.na(EVENT_TYPE))
#joined_data

# Remove rows with no species data

joined_data <- joined_data %>%
  filter(specificEpithet != "")
joined_data <- joined_data %>%
  filter(specificEpithet != " ")

```

Data Analysis & Hypothesis Testing - AB Test for Storm vs No Storm

Before we do our AB test to determine whether extreme weather impacts bee diversity, we must create a null hypothesis to either accept or reject. Our null hypothesis will be: Bees are equally diverse, according to the Shannon diversity index, in areas which have recently been impacted by either extreme heat or freezing temperatures. If we get a p value of less than 0.05, we will reject this null and assume that the weather does impact bees.

To do our analysis, we will create a function to compute Shannon Diversity, then compute our observed statistic for Shannon diversity of bees collected after certain extreme weather events. Our observed statistic will be the difference in Shannon Diversities between bees collected outside of the storm timeframe (30 days) and bees collected within 30 days of a storm. After that, we will collect some arbitrary number (`n_iter = 1000`) to iterate through. In each iteration, we will shuffle the control column and re-compute the differences. The results will show if our observed statistic falls within the 95% confidence interval of the sampled statistics.

```

# Find Species Richness
S <- length(unique(joined_data$specificEpithet))
S

## [1] 467

# Natural Log of Species Richness (Necessary for Species evenness, but not Shannon Diversity)
ln_s <- log(S)
ln_s

## [1] 6.146329

```

```

# Calculate Proportion of Each Species

species_counts <- joined_data %>%
  count(specificEpithet, name = 'n')

proportions <- c()

j <- 1

for (i in species_counts$n) {
  proportions[j] = (i / sum(species_counts$n))
  j <- j+1
}
#proportions

```

```

# Calculate Overall Shannon Diversity Index
#  $H = - \text{Summation}[P(i) * \ln P(i)]$ .

summation <- 0
for (p in proportions) {
  summation <- summation + (p * log(p))
}

shannon_diversity_index <- (summation * -1)

```

There is already a package to compute Shannon diversity, however we created a reproducible function to help us better understand how it runs as well as comparing the results of the two functions to make sure our numbers line up.

```

# Calculate Shannon Diversity Index Another Way

library(vegan)

```

```
## Loading required package: permute
```

```

shannon_diversity_index_2 <- joined_data %>%
  summarize(diversity = diversity(table(specificEpithet)))
shannon_diversity_index_2

```

```

##    diversity
## 1  4.650002

```

```

# Calculate Shannon Diversity for Control Group

joined_data
control_table <- joined_data %>%
  filter(Control == TRUE)

S_control <- length(unique(control_table$specificEpithet))

species_counts_control <- control_table %>%

```

```

count(specificEpithet, name = 'n')

proportions_control <- c()

j <- 1

for (i in species_counts_control$n) {
  proportions_control[j] = (i / sum(species_counts_control$n))
  j <- j+1
}
proportions_control

summation_control <- 0
for (p in proportions_control) {
  summation_control <- summation_control + (p * log(p))
}

shannon_diversity_index_control <- (summation_control * -1)

shannon_diversity_index_control

```

```

# Calculate Shannon Diversity of Treatment Group
joined_data
not_control_table <- joined_data %>%
  filter(Control == FALSE)
control_table <- joined_data %>%
  filter(Control == TRUE)

S_not_control <- length(unique(not_control_table$specificEpithet))

species_counts_not_control <- not_control_table %>%
  count(specificEpithet, name = 'n')

proportions_not_control <- c()

j <- 1

for (i in species_counts_not_control$n) {
  proportions_not_control[j] = (i / sum(species_counts_not_control$n))
  j <- j+1
}
proportions_not_control

summation_not_control <- 0
for (p in proportions_not_control) {
  summation_not_control <- summation_not_control + (p * log(p))
}

shannon_diversity_index_treatment <- (summation_not_control * -1)

shannon_diversity_index_treatment

```

```
# Create function to compute Shannon Diversity (reproducible)
```

```
shannon_diversity <- function(df, col) {  
  col <- deparse(substitute(col))  
  
  S <- length(unique(df[[col]]))  
  
  species_counts <- df %>%  
    count(.data[[col]], name = 'n')  
  
  proportions <- c()  
  
  j <- 1  
  
  for (i in species_counts$n) {  
    proportions[j] = (i / sum(species_counts$n))  
    j <- j+1  
  }  
  
  summation <- 0  
  for (p in proportions) {  
    summation <- summation + (p * log(p))  
  }  
  
  H <- (summation * -1)  
  
  return(H)  
}  
  
shannon_diversity(not_control_table, specificEpithet)
```

```
## [1] 4.385279
```

The Shannon diversity for “extreme weather event bees” was 4.3852, and the Shannon diversity for bees captured in normal circumstances (Control) was 4.646. The diversity is lower for bees captured following events (Extreme heat or freezing temperatures), but is it significantly so? To determine this we will use A/B testing.

```
# Set seed for reproducibility, and the number of iterations to shuffle  
set.seed(1)  
n_iterations <- 1000
```

```
# Perform AB Testing
```

```
shuffle_results <- c()  
joined_data_filtered_true <- joined_data %>%  
  filter(Control == TRUE)  
joined_data_filtered_false <- joined_data %>%  
  filter(Control == FALSE)  
  
for (i in 1:n_iterations) {
```

```

joined_data_sampled <- joined_data %>%
  mutate(shuffled = sample(Control))
temp <- joined_data_sampled %>%
  filter(shuffled == TRUE)
temp2 <- joined_data_sampled %>%
  filter(shuffled == FALSE)
shuffle_results[i] <- shannon_diversity(temp, specificEpithet) - shannon_diversity(temp2, specificEpithet)
}

sort(shuffle_results)

```

```

# Assign P Value (Probability that our observed shannon diversity difference between storm/no storm for
p_value <- mean(shuffle_results <= shannon_diversity(not_control_table, specificEpithet) - shannon_diversity(not_control_table, specificEpithet))
p_value

```

```
## [1] 0
```

A P value of 0 means that we reject the null hypothesis. This suggests that extreme weather events do impact bee diversity, making the bees less diverse in areas that recently experienced either extreme heat or freezing temperatures.

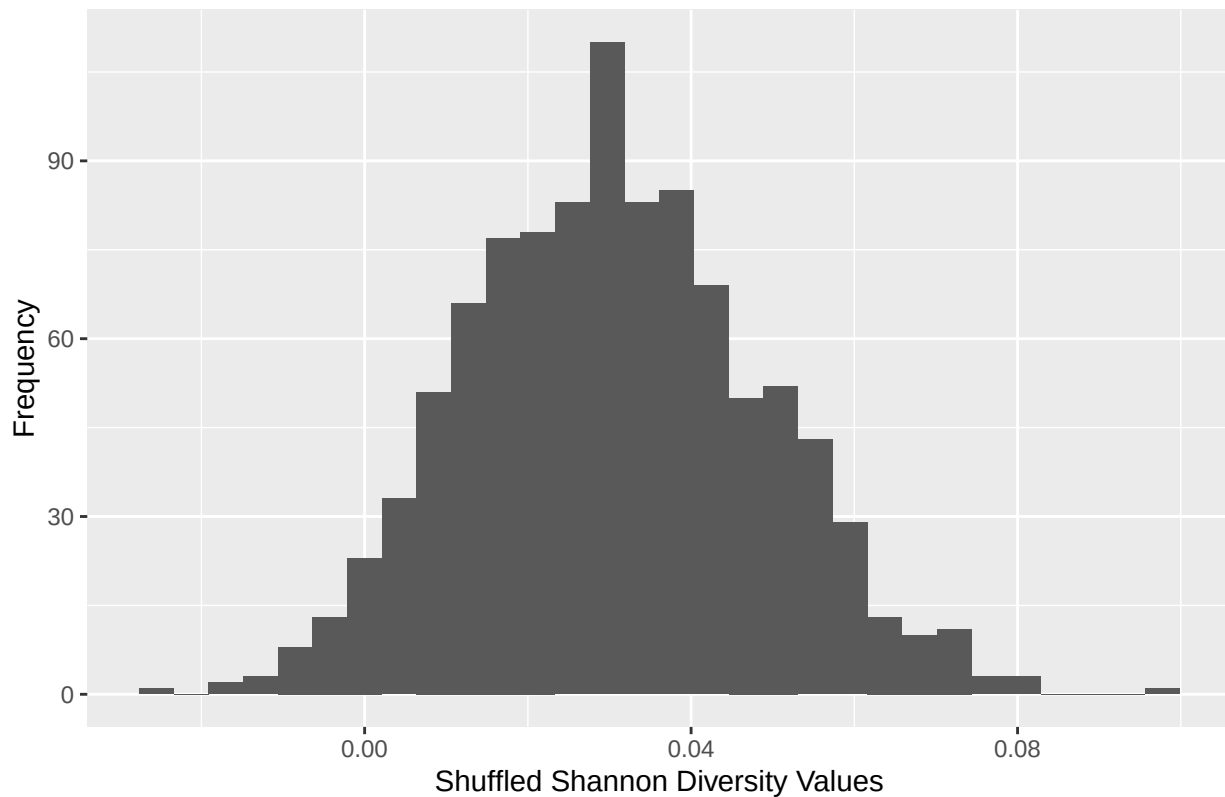
```

# Histogram of Shuffled Results
ggplot(data.frame(x = shuffle_results), aes(x)) +
  geom_histogram() +
  ggtitle("Distribution of Shuffled AB Test Shannon Diversities") +
  xlab("Shuffled Shannon Diversity Values") +
  ylab("Frequency")

```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Distribution of Shuffled AB Test Shannon Diversities

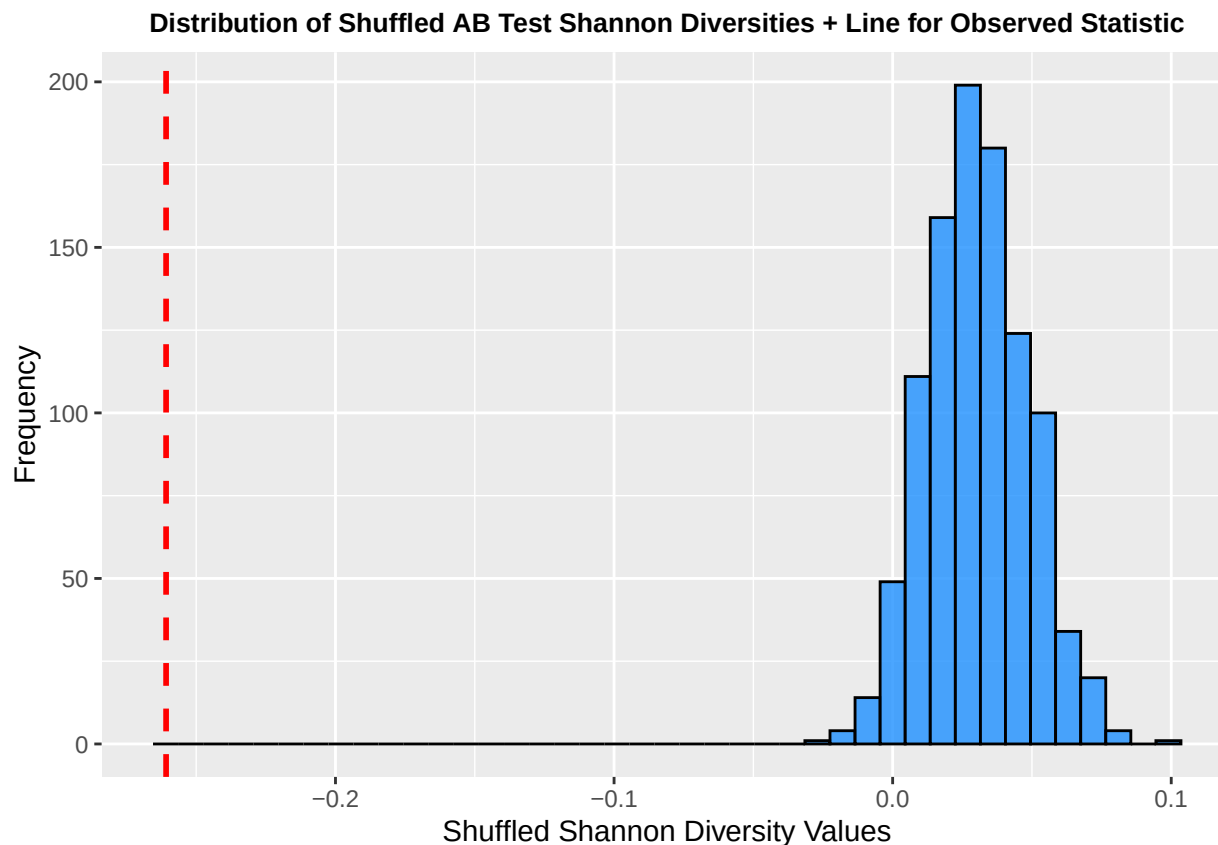


```
shannon_diversity(joined_data_filtered_true, specificEpithet) - shannon_diversity(joined_data_filtered_
```

```
## [1] 0.2608057
```

```
# Histogram of Shuffled Results, plus vertical line showing non-shuffled diversity
ggplot(data.frame(x = shuffle_results), aes(x)) +
  geom_histogram(fill = "#1E90FF", color = "black", alpha = 0.8, binwidth = 0.009) +
  ggtitle("Distribution of Shuffled AB Test Shannon Diversities + Line for Observed Statistic") +
  geom_vline(xintercept = -(shannon_diversity(joined_data_filtered_true, specificEpithet) - shannon_div
  xlab("Shuffled Shannon Diversity Values") +
  ylab("Frequency") +
  theme(plot.title = element_text(size = 10, face = "bold", hjust = 0.5))
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



As we can see, our observed statistic (Shannon diversity of storm bees) falls well outside of the limits of our shuffled results, meaning it is statistically significant.

Data Analysis: Does freezing temperatures impact bee diversity?

Now, we want to do another AB test, but this time focused on frost/freeze impacts on bee diversity (ignoring extreme heat). Our null hypothesis this time will be: Freezing temperatures do not impact bee diversity according to the Shannon diversity index.

```
# Create new table to analyze freezing temperatures
```

```
joined_data_frost <- joined_data %>%
  mutate(frost = coalesce(EVENT_TYPE == "Frost/Freeze", FALSE))
joined_data_frost
```

```
# Calculate Observed Statistic
```

```
joined_data_frost_filtered_true <- joined_data_frost %>%
  filter(frost == TRUE)
joined_data_frost_filtered_false <- joined_data_frost %>%
  filter(frost == FALSE)
shannon_diversity(joined_data_frost_filtered_true, specificEpithet) - shannon_diversity(joined_data_frost_filtered_false, specificEpithet)
```

```
## [1] -0.2722803
```

```

# Perform AB Test
shuffled_frost_results <- c()
for (i in 1:n_iterations) {
  joined_data_frost <- joined_data_frost %>%
    mutate(frost_shuffled = sample(frost))
  temp <- joined_data_frost %>%
    filter(frost_shuffled == TRUE)
  temp2 <- joined_data_frost %>%
    filter(frost_shuffled == FALSE)
  shuffled_frost_results[i] <- shannon_diversity(temp, specificEpithet) - shannon_diversity(temp2, specificEpithet)
}

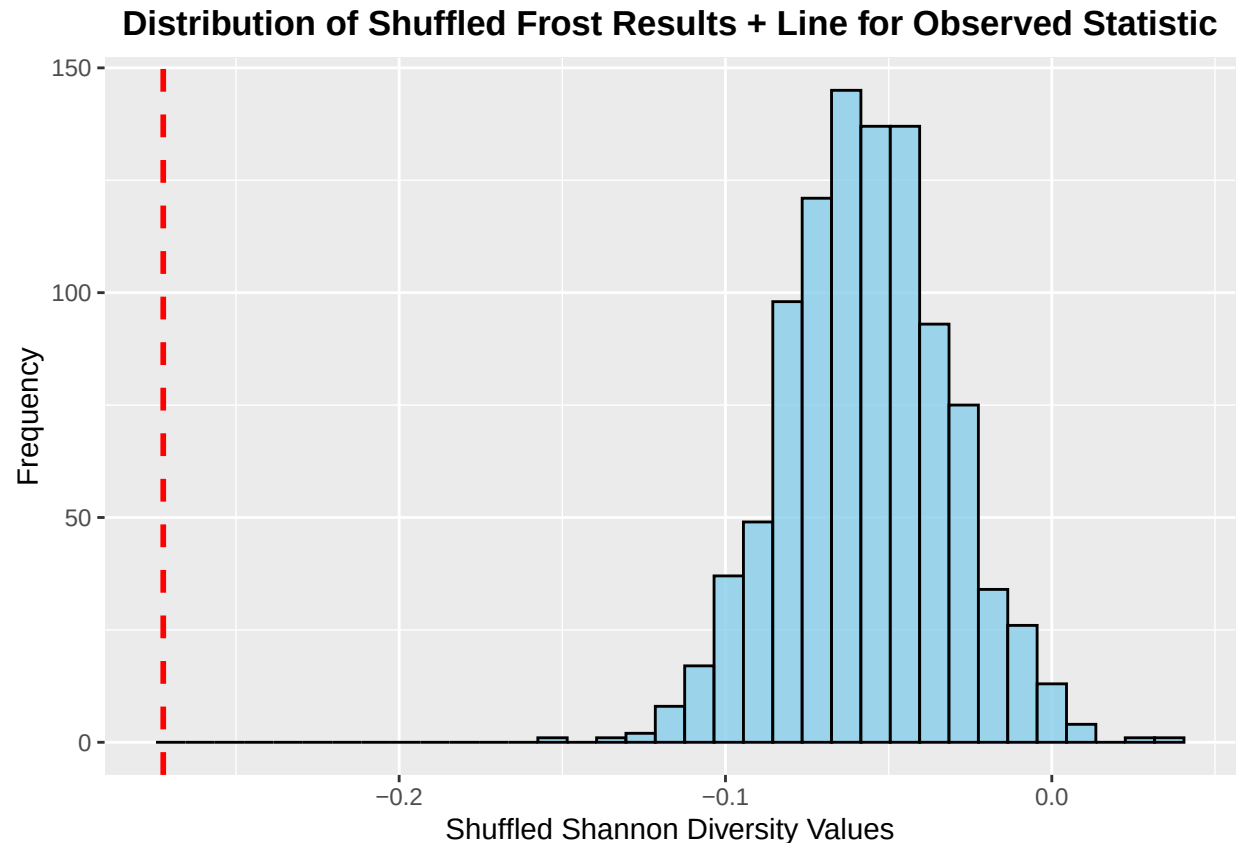
sort(shuffled_frost_results)

# Compute P Value for freezing temperatures
p_value_frost <- mean(shuffled_frost_results <= shannon_diversity(joined_data_frost_filtered_true, specificEpithet) - shannon_diversity(joined_data_frost_filtered_false, specificEpithet))
p_value_frost

## [1] 0

# Histogram for frost results
ggplot(data.frame(x = shuffled_frost_results), aes(x)) +
  geom_histogram(fill = "skyblue", color = "black", alpha = 0.8, binwidth = 0.009) +
  ggtitle("Distribution of Shuffled Frost Results + Line for Observed Statistic") +
  geom_vline(xintercept = shannon_diversity(joined_data_frost_filtered_true, specificEpithet) - shannon_diversity(joined_data_frost_filtered_false, specificEpithet)) +
  xlab("Shuffled Shannon Diversity Values") +
  ylab("Frequency") +
  theme(plot.title = element_text(size = 13, face = "bold", hjust = 0.5))

```



As we see here, the p value is once again 0. This means that we reject the null, which suggests that freezing temperatures does negatively impact the Shannon diversity of bees.

Data Analysis - AB Test for Extreme Heat

For our last AB test, we will focus on how extreme heat impacts bee diversity. Our null hypothesis: Extreme heat events do not impact the diversity of bees.

```
# Create new table to analyze extreme heat
```

```
joined_data_heat <- joined_data %>%
  mutate(heat = coalesce(EVENT_TYPE == "Excessive Heat", FALSE))
joined_data_heat
```

```
# Calculate Observed Statistic
```

```
joined_data_heat_filtered_true <- joined_data_heat %>%
  filter(heat == TRUE)
joined_data_heat_filtered_false <- joined_data_heat %>%
  filter(heat == FALSE)
shannon_diversity(joined_data_heat_filtered_true, specificEpithet) - shannon_diversity(joined_data_heat_filtered_false, specificEpithet)
```

```
## [1] -0.7953634
```

```

# Perform AB Test
shuffled_heat_results <- c()
for (i in 1:n_iterations) {
  joined_data_heat_temp <- joined_data_heat %>%
    mutate(heat_shuffled = sample(heat))
  temp <- joined_data_heat_temp %>%
    filter(heat_shuffled == TRUE)
  temp2 <- joined_data_heat_temp %>%
    filter(heat_shuffled == FALSE)
  shuffled_heat_results[i] <- shannon_diversity(temp, specificEpithet) - shannon_diversity(temp2, specificEpithet)
}

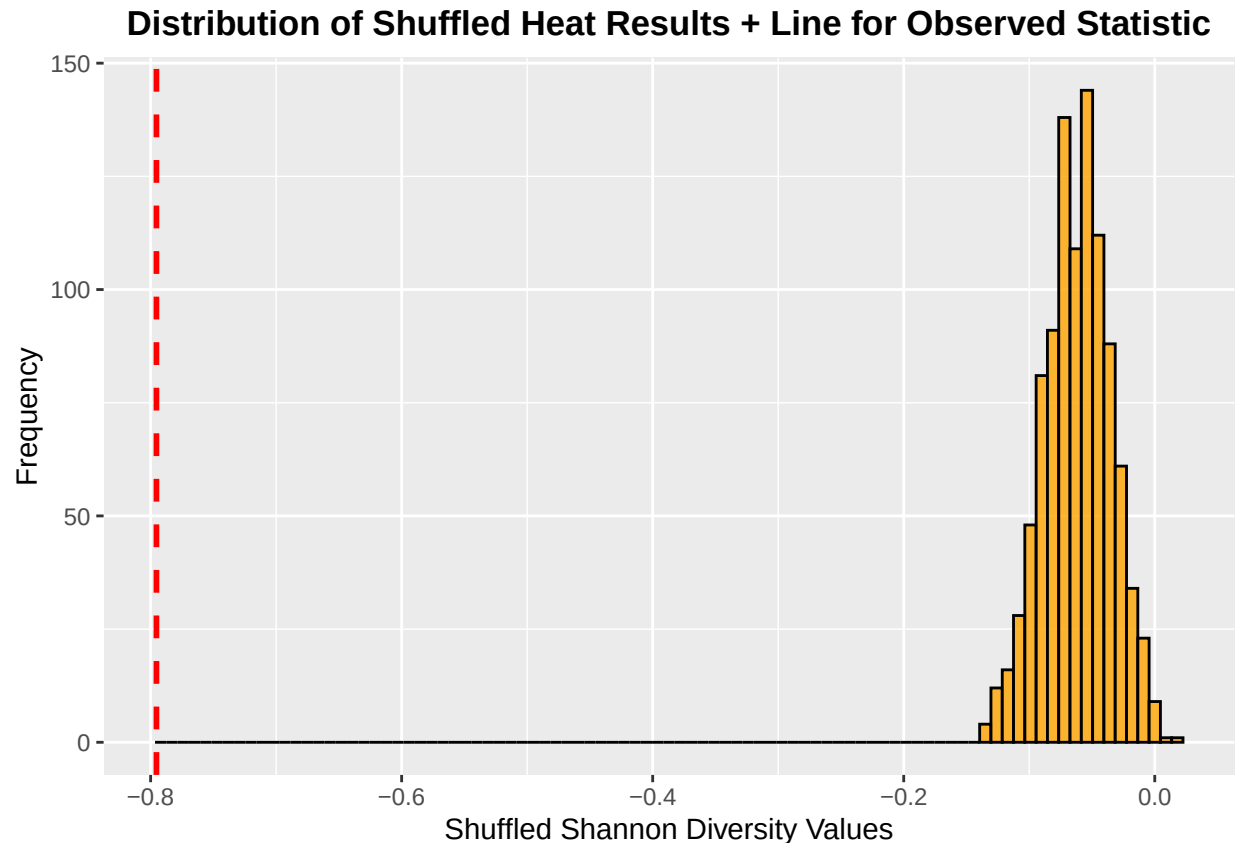
sort(shuffled_heat_results)

# Compute P Value for Shannon Diversity of Heat Bees
p_value_heat <- mean(shuffled_heat_results <= shannon_diversity(joined_data_heat_filtered_true, specificEpithet) - shannon_diversity(joined_data_heat_filtered_false, specificEpithet))
p_value_heat

## [1] 0

# Histogram for Sampled Heat Shannon Diversity Differences
ggplot(data.frame(x = shuffled_heat_results), aes(x)) +
  geom_histogram(fill = "orange", color = "black", alpha = 0.8, binwidth = 0.009) +
  ggtitle("Distribution of Shuffled Heat Results + Line for Observed Statistic") +
  geom_vline(xintercept = shannon_diversity(joined_data_heat_filtered_true, specificEpithet) - shannon_diversity(joined_data_heat_filtered_false, specificEpithet)) +
  xlab("Shuffled Shannon Diversity Values") +
  ylab("Frequency") +
  theme(plot.title = element_text(size = 13, face = "bold", hjust = 0.5))

```



Once again, we reject the Null hypothesis. This suggests that extreme heat has a significant impact on bee diversity. If you compare the difference extreme heat and cold AB for Shannon diversities, heat appears to have a greater impact on diversity than extreme cold!

Visualizations

The following figures show how diversity and community structure are affected by the sampled extreme weather events.

Effect on Shannon diversity

To visualize how diversity is affected by the sampled weather events we first had to find the dates of our sampled weather events. Once dates of events were found we used a for loop to subset the data to samples within 30 day windows before and after events and then apply a Shannon diversity function to the subsetted data to give us a data frame that listed Shannon diversity values before and after events. For ease of visualization, we took the mean of these diversity values per event type and plotted them on a bar chart. We chose a bar plot because it's the clearest way to compare diversity values before and after events by event type. Visualization shows that there is a noticeable difference in Shannon diversity before and after events with lower values following extreme heat and cold events. This indicates that extreme weather events lead to a decrease in bee diversity across Oregon following these events.

```
# Find dates of events
Event_Date_Data <- joined_data %>%
  filter(!is.na(EVENT_TYPE)) %>%
```

```

select(BEGIN_DATE, EVENT_TYPE) %>%
distinct()

# Create subsets of bees 30-days pre/post events and compute Shannon diversity for each
n_events <- nrow(Event_Date_Data)

pre_div <- numeric(n_events)
post_div <- numeric(n_events)

for (i in 1:n_events) {
  event_day <- Event_Date_Data$BEGIN_DATE[i]

  # Bees 30 days before
  pre_bees <- joined_data %>%
    filter(verbatimEventDate >= event_day - 30,
           verbatimEventDate < event_day)

  # Bees 30 days after
  post_bees <- joined_data %>%
    filter(verbatimEventDate > event_day,
           verbatimEventDate <= event_day + 30)

  # Counts per species
  pre_tab <- table(pre_bees$GenusSpecies)
  post_tab <- table(post_bees$GenusSpecies)

  # Calculate Shannon diversity
  pre_div[i] <- diversity(pre_tab)
  post_div[i] <- diversity(post_tab)
}

# Comparison Table
diversity_results <- data.frame(EVENT_DATE = Event_Date_Data$BEGIN_DATE,
                                EVENT_TYPE = Event_Date_Data$EVENT_TYPE,
                                shannon_pre30 = pre_div,
                                shannon_post30 = post_div,
                                diff_post_minus_pre = post_div - pre_div)

diversity_results

```

##	EVENT_DATE	EVENT_TYPE	shannon_pre30	shannon_post30	diff_post_minus_pre
## 1	2018-06-01	Frost/Freeze	4.156443	4.364632	0.20818872
## 2	2019-04-28	Frost/Freeze	3.724068	4.372787	0.64871958
## 3	2019-05-01	Frost/Freeze	3.842176	4.439451	0.59727449
## 4	2019-05-02	Frost/Freeze	3.853184	4.451249	0.59806470
## 5	2019-05-03	Frost/Freeze	3.865729	4.463131	0.59740155
## 6	2019-06-06	Frost/Freeze	4.453491	4.544260	0.09076955
## 7	2019-06-08	Frost/Freeze	4.478794	4.547855	0.06906015
## 8	2019-06-21	Frost/Freeze	4.550445	4.498289	-0.05215627
## 9	2019-08-27	Excessive Heat	4.137750	3.774997	-0.36275241
## 10	2019-10-09	Frost/Freeze	3.646936	1.264468	-2.38246771
## 11	2019-10-10	Frost/Freeze	3.646374	1.106318	-2.54005601
## 12	2019-10-11	Frost/Freeze	3.636160	1.106318	-2.52984219

```
## 13 2019-10-12 Frost/Freeze 3.665144 1.242974 -2.42216976
## 14 2020-04-01 Frost/Freeze 3.279817 3.823195 0.54337781
## 15 2020-04-02 Frost/Freeze 3.313106 3.830059 0.51695366
## 16 2020-04-13 Frost/Freeze 3.491576 3.958374 0.46679817
## 17 2020-04-14 Frost/Freeze 3.478238 3.965962 0.48772405
## 18 2020-06-08 Frost/Freeze 4.222775 4.614766 0.39199111
## 19 2020-06-17 Frost/Freeze 4.294349 4.648998 0.35464854
## 20 2020-08-14 Excessive Heat 4.253941 3.626695 -0.62724574
## 21 2020-09-06 Excessive Heat 3.885564 3.079177 -0.80638680
## 22 2020-10-22 Frost/Freeze 3.008084 1.580819 -1.42726524
## 23 2020-10-23 Frost/Freeze 2.993515 1.580819 -1.41269665
## 24 2020-10-25 Frost/Freeze 3.003195 1.580819 -1.42237615
## 25 2020-10-26 Frost/Freeze 3.003195 1.580819 -1.42237615
## 26 2021-03-30 Frost/Freeze 3.283154 3.769597 0.48644280
## 27 2021-03-31 Frost/Freeze 3.292844 3.771198 0.47835365
## 28 2021-04-06 Frost/Freeze 3.239229 3.848569 0.60933986
## 29 2021-04-07 Frost/Freeze 3.246744 3.860849 0.61410466
## 30 2021-04-09 Frost/Freeze 3.313100 3.883203 0.57010289
## 31 2021-04-11 Frost/Freeze 3.249293 3.919341 0.67004818
## 32 2021-05-08 Frost/Freeze 3.860849 4.166085 0.30523632
## 33 2021-02-25 Winter Storm 1.410514 3.283154 1.87264028
## 34 2021-06-26 Excessive Heat 4.408524 4.118752 -0.28977251
## 35 2021-06-20 Excessive Heat 4.321547 4.285652 -0.03589543
## 36 2021-06-27 Excessive Heat 4.408514 4.107565 -0.30094904
## 37 2021-06-25 Excessive Heat 4.421981 4.132673 -0.28930786
## 38 2021-07-29 Excessive Heat 4.102714 3.201618 -0.90109616
## 39 2021-06-07 Frost/Freeze 4.159186 4.458887 0.29970087
## 40 2021-08-10 Excessive Heat 3.764718 2.967495 -0.79722345
## 41 2021-08-11 Excessive Heat 3.723790 2.924247 -0.79954301
## 42 2021-08-14 Excessive Heat 3.652266 3.050090 -0.60217609
## 43 2018-06-10 Frost/Freeze 4.256592 4.359671 0.10307889
## 44 2018-04-03 Frost/Freeze 2.781516 3.552491 0.77097576
## 45 2018-09-12 Frost/Freeze 3.740318 3.171988 -0.56833088
## 46 2018-09-14 Frost/Freeze 3.697300 3.131537 -0.56576235
## 47 2018-09-15 Frost/Freeze 3.705795 3.136244 -0.56955100
```

```
# Find the means for each event type
event_type_summary <- diversity_results %>%
  group_by(EVENT_TYPE) %>%
  summarise(mean_pre = mean(shannon_pre30),
            mean_post = mean(shannon_post30),
            mean_difference = mean(diff_post_minus_pre),
            n_events = n())

event_type_summary
```

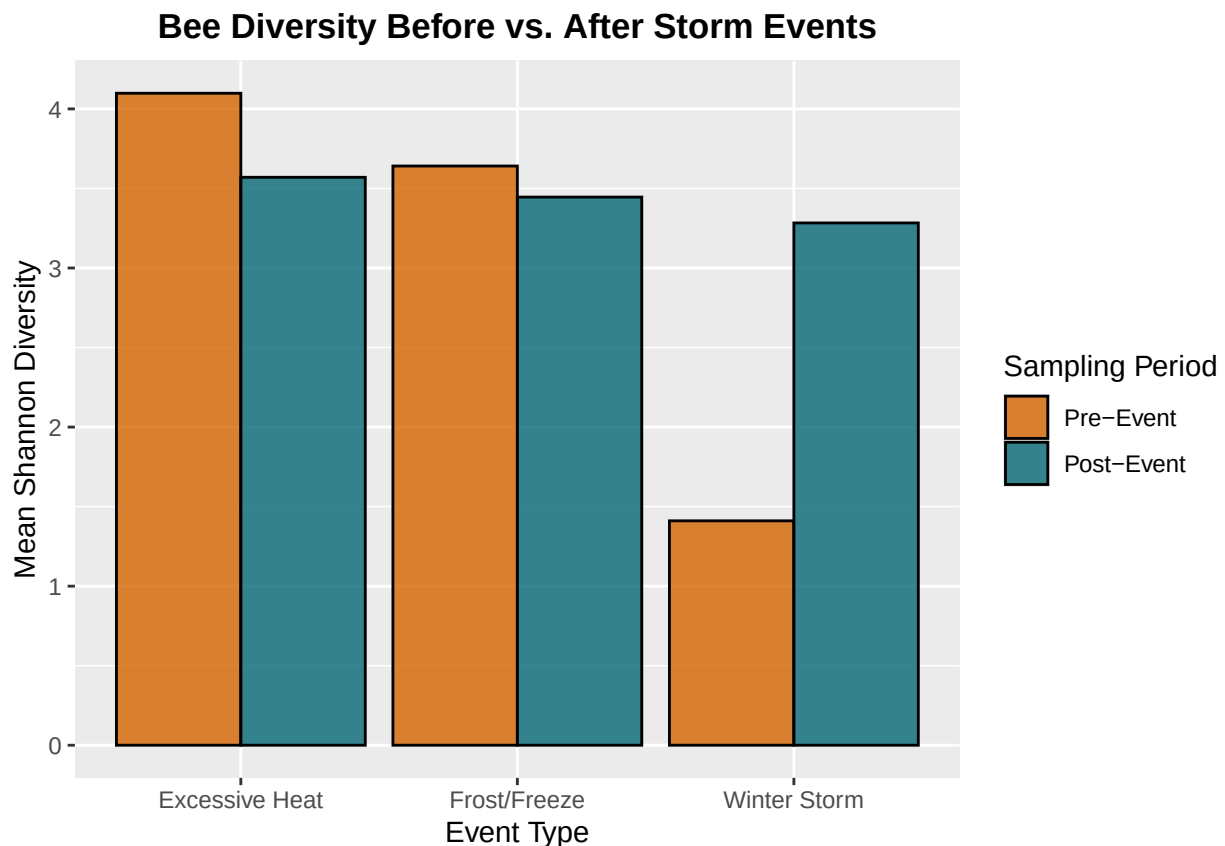
```
## # A tibble: 3 x 5
##   EVENT_TYPE    mean_pre mean_post mean_difference n_events
##   <chr>         <dbl>    <dbl>         <dbl>    <int>
## 1 Excessive Heat    4.10      3.57         -0.528      11
## 2 Frost/Freeze     3.64      3.45         -0.195      35
## 3 Winter Storm     1.41      3.28          1.87        1
```

```

# Bar graph showing differences in pre/post event, grouped by event type
# Reshape data for plotting
plot_data <- event_type_summary %>%
  select(EVENT_TYPE, mean_pre, mean_post) %>%
  pivot_longer(cols = c(mean_pre, mean_post), names_to = "period", values_to = "diversity")
# Change order of data for plot
plot_data$period <- factor(plot_data$period,
                           levels = c("mean_pre", "mean_post"),
                           labels = c("Pre-Event", "Post-Event"))

# Plot data
ggplot(plot_data, aes(x = EVENT_TYPE, y = diversity, fill = period)) +
  geom_col(position = "dodge", color = "black", alpha = 0.8) +
  ggtitle("Bee Diversity Before vs. After Storm Events") +
  xlab("Event Type") +
  ylab("Mean Shannon Diversity") +
  theme(plot.title = element_text(size = 13, face = "bold", hjust = 0.5)) +
  #theme_minimal() +
  labs(fill = "Sampling Period") +
  scale_fill_manual(values=c("#D46600", "#056875"))

```



Effect on community structure

To compare community structure before and after extreme weather events bees were classified as specialists or generalists using genus-level traits derived from established ecological references (Michener (2000); Wilson

and Carril (2015); Danforth et al. (2019)). Genera with strong evidence of specialization (e.g., *Diadasia*, *Peponapis*, *Perdita*, *Chelostoma*, *Dufourea*) were categorized as specialists based on associations documented by Fowler (2016) and Praz, Müller, and Dorn (2018) and genera (e.g., *Bombus*, *Lasioglossum*, *Halictus*) were categorized as generalists through documentations by Wilson and Carril (2015) and Goulson (2010). Bee identities were also cross-referenced with the USGS Native Bee Inventory dataset (Fowler (2016)), which is a widely used North American bee monitoring program. Genera were only assigned to specialist or generalist categories when consistent agreement existed in the literature. Ambiguous genera were excluded from assignment to avoid confusion or bias in the data. We decided to use genera instead of species-specific traits due to time constraints, complexity, and significant disagreement in the literature. Once genera were categorized we created a new data table and used a for loop to subset the data for bee samples within 30 day windows before and after events, count the number of samples for each genera, and then find the overall proportions relative to specialists for ease of plotting. Once this was completed we summarized the data by event type and plotted the data in a bar chart. The bar chart shows that there is a shift in community structure towards generalist genera which indicates that extreme weather events cause a noticeable difference in community structure with specialist genera not able to survive as well as generalist species after these events.

```
# Create object vectors for Genera dependent on if they are specialists or generalists. Genera that have
specialist_genera <- c("Diadasia", "Peponapis", "Hesperapis", "Dufourea", "Chelostoma", "Perdita", "Colletes")
generalist_genera <- c("Bombus", "Halictus", "Apis", "Lasioglossum", "Eucera", "Ceratina", "Agapostemon",
                      "Xylocopa", "Osmia", "Anthophora", "Megachile", "Hylaeus", "Calliopsis", "Heriades",
                      "Ashmeadiella", "Atoposmia", "Trachusa", "Dianthidium", "Tetraloniella",
                      "Pseudoanthidium", "Protosmia", "Anthidiellum", "Hoplitis", "Habropoda")

# For clarity
excluded_genera <- c("Andrena", "Micralictoides", "Stelis", "Sphecodes", "Melecta", "Coelioxys", "Epeolus",
                    "Neolarra", "Oreopasites", "Nomia", "Triepeolus", "Biastes", "Epimelissodes",
                    "Protohalonia", "Dioxys", "Zacosmia", "Holcopasites", "Brachymelecta")

# Make a new data frame and loop to analyze community structure
# Create data table
structure_table <- joined_data %>%
  filter(!is.na(EVENT_TYPE)) %>%
  select(BEGIN_DATE, EVENT_TYPE) %>%
  distinct()
n_events_s <- nrow(structure_table)

# Create Vectors to store info from for loop
pre_spec      <- numeric(n_events)
pre_gen       <- numeric(n_events)
post_spec     <- numeric(n_events)
post_gen      <- numeric(n_events)
pre_prop_spec <- numeric(n_events)
post_prop_spec <- numeric(n_events)

# For loop to find proportions of generalist vs specialist bees and how extreme weather affected them
for (i in 1:n_events_s) {

  event_day <- structure_table$BEGIN_DATE[i]

  # Bees 30 days before event
```

```

pre_beets <- joined_data %>%
  filter(verbatimEventDate >= event_day - 30,
         verbatimEventDate < event_day) %>%
  filter(genus %in% c(specialist_genera, generalist_genera))

# Bees 30 days after event
post_beets <- joined_data %>%
  filter(verbatimEventDate > event_day,
         verbatimEventDate <= event_day + 30) %>%
  filter(genus %in% c(specialist_genera, generalist_genera))

pre_spec[i] <- sum(pre_beets$genus %in% specialist_genera)
pre_gen[i] <- sum(pre_beets$genus %in% generalist_genera)
post_spec[i] <- sum(post_beets$genus %in% specialist_genera)
post_gen[i] <- sum(post_beets$genus %in% generalist_genera)

pre_total <- pre_spec[i] + pre_gen[i]
post_total <- post_spec[i] + post_gen[i]

# Found proportions kept giving 0 values so need to make sure that we aren't dividing by 0
if (pre_total > 0) {
  pre_prop_spec[i] <- pre_spec[i] / pre_total
} else {
  pre_prop_spec[i] <- NA
}
if (post_total > 0) {
  post_prop_spec[i] <- post_spec[i] / post_total
} else {
  post_prop_spec[i] <- NA
}
}

```

```

# Comparison table
structure_results <- data.frame(EVENT_DATE = structure_table$BEGIN_DATE,
                                EVENT_TYPE = structure_table$EVENT_TYPE,
                                pre_spec = pre_spec,
                                pre_gen = pre_gen,
                                post_spec = post_spec,
                                post_gen = post_gen,
                                pre_prop_spec = pre_prop_spec,
                                post_prop_spec = post_prop_spec,
                                diff_prop_spec = post_prop_spec - pre_prop_spec)

structure_results

```

##	EVENT_DATE	EVENT_TYPE	pre_spec	pre_gen	post_spec	post_gen	pre_prop_spec
## 1	2018-06-01	Frost/Freeze	9	1763	35	1542	0.0050790068
## 2	2019-04-28	Frost/Freeze	1	1347	43	3265	0.0007418398
## 3	2019-05-01	Frost/Freeze	0	1698	50	3222	0.0000000000
## 4	2019-05-02	Frost/Freeze	0	1753	55	3257	0.0000000000
## 5	2019-05-03	Frost/Freeze	0	1772	55	3172	0.0000000000
## 6	2019-06-06	Frost/Freeze	55	3442	177	3994	0.0157277667
## 7	2019-06-08	Frost/Freeze	54	3038	209	4129	0.0174644243

## 8	2019-06-21	Frost/Freeze	70	3916	287	3156	0.0175614651
## 9	2019-08-27	Excessive Heat	347	3385	124	1180	0.0929796356
## 10	2019-10-09	Frost/Freeze	80	783	0	44	0.0926998841
## 11	2019-10-10	Frost/Freeze	80	784	0	41	0.0925925926
## 12	2019-10-11	Frost/Freeze	80	776	0	41	0.0934579439
## 13	2019-10-12	Frost/Freeze	80	741	0	43	0.0974421437
## 14	2020-04-01	Frost/Freeze	0	168	0	1568	0.0000000000
## 15	2020-04-02	Frost/Freeze	0	173	0	1581	0.0000000000
## 16	2020-04-13	Frost/Freeze	0	429	0	2045	0.0000000000
## 17	2020-04-14	Frost/Freeze	0	500	0	1998	0.0000000000
## 18	2020-06-08	Frost/Freeze	28	2700	139	2740	0.0102639296
## 19	2020-06-17	Frost/Freeze	110	3067	142	3147	0.0346238590
## 20	2020-08-14	Excessive Heat	204	2974	79	1505	0.0641913153
## 21	2020-09-06	Excessive Heat	84	2322	37	508	0.0349127182
## 22	2020-10-22	Frost/Freeze	23	387	0	13	0.0560975610
## 23	2020-10-23	Frost/Freeze	21	386	0	13	0.0515970516
## 24	2020-10-25	Frost/Freeze	21	375	0	13	0.0530303030
## 25	2020-10-26	Frost/Freeze	21	375	0	13	0.0530303030
## 26	2021-03-30	Frost/Freeze	0	66	0	1491	0.0000000000
## 27	2021-03-31	Frost/Freeze	0	74	0	1643	0.0000000000
## 28	2021-04-06	Frost/Freeze	0	153	6	1937	0.0000000000
## 29	2021-04-07	Frost/Freeze	0	174	6	1968	0.0000000000
## 30	2021-04-09	Frost/Freeze	0	184	6	2036	0.0000000000
## 31	2021-04-11	Frost/Freeze	0	225	6	2296	0.0000000000
## 32	2021-05-08	Frost/Freeze	6	1968	13	1993	0.0030395137
## 33	2021-02-25	Winter Storm	0	13	0	66	0.0000000000
## 34	2021-06-26	Excessive Heat	50	2617	261	2823	0.0187476565
## 35	2021-06-20	Excessive Heat	48	2349	160	2764	0.0200250313
## 36	2021-06-27	Excessive Heat	53	2629	258	2727	0.0197613721
## 37	2021-06-25	Excessive Heat	53	2761	253	2776	0.0188343994
## 38	2021-07-29	Excessive Heat	260	2684	78	1323	0.0883152174
## 39	2021-06-07	Frost/Freeze	13	2058	106	3141	0.0062771608
## 40	2021-08-10	Excessive Heat	262	2094	27	1292	0.1112054329
## 41	2021-08-11	Excessive Heat	259	1939	27	1271	0.1178343949
## 42	2021-08-14	Excessive Heat	231	1768	37	1211	0.1155577789
## 43	2018-06-10	Frost/Freeze	9	1783	55	1589	0.0050223214
## 44	2018-04-03	Frost/Freeze	0	37	0	423	0.0000000000
## 45	2018-09-12	Frost/Freeze	194	1846	67	345	0.0950980392
## 46	2018-09-14	Frost/Freeze	217	1819	37	246	0.1065815324
## 47	2018-09-15	Frost/Freeze	222	1787	37	249	0.1105027377
##	post_prop_spec diff_prop_spec						
## 1	0.022194039	0.017115033					
## 2	0.012998791	0.012256951					
## 3	0.015281174	0.015281174					
## 4	0.016606280	0.016606280					
## 5	0.017043694	0.017043694					
## 6	0.042435867	0.026708100					
## 7	0.048178884	0.030714460					
## 8	0.083357537	0.065796072					
## 9	0.095092025	0.002112389					
## 10	0.000000000	-0.092699884					
## 11	0.000000000	-0.092592593					
## 12	0.000000000	-0.093457944					
## 13	0.000000000	-0.097442144					

```
## 14    0.000000000    0.000000000
## 15    0.000000000    0.000000000
## 16    0.000000000    0.000000000
## 17    0.000000000    0.000000000
## 18    0.048280653    0.038016723
## 19    0.043174217    0.008550358
## 20    0.049873737   -0.014317578
## 21    0.067889908    0.032977190
## 22    0.000000000   -0.056097561
## 23    0.000000000   -0.051597052
## 24    0.000000000   -0.053030303
## 25    0.000000000   -0.053030303
## 26    0.000000000    0.000000000
## 27    0.000000000    0.000000000
## 28    0.003088008    0.003088008
## 29    0.003039514    0.003039514
## 30    0.002938296    0.002938296
## 31    0.002606429    0.002606429
## 32    0.006480558    0.003441045
## 33    0.000000000    0.000000000
## 34    0.084630350    0.065882694
## 35    0.054719562    0.034694531
## 36    0.086432161    0.066670789
## 37    0.083525916    0.064691517
## 38    0.055674518   -0.032640699
## 39    0.032645519    0.026368358
## 40    0.020470053   -0.090735380
## 41    0.020801233   -0.097033162
## 42    0.029647436   -0.085910343
## 43    0.033454988    0.028432666
## 44    0.000000000    0.000000000
## 45    0.162621359    0.067523320
## 46    0.130742049    0.024160517
## 47    0.129370629    0.018867892
```

```
# Plot the data
# Summarize by event type
structure_summary <- structure_results %>%
  group_by(EVENT_TYPE) %>%
  summarise(mean_pre_prop_spec = mean(pre_prop_spec, na.rm = TRUE),
            mean_post_prop_spec = mean(post_prop_spec, na.rm = TRUE))

structure_plot_data <- structure_summary %>%
  pivot_longer(cols = c(mean_pre_prop_spec, mean_post_prop_spec), names_to = "period", values_to = "prop_spec")

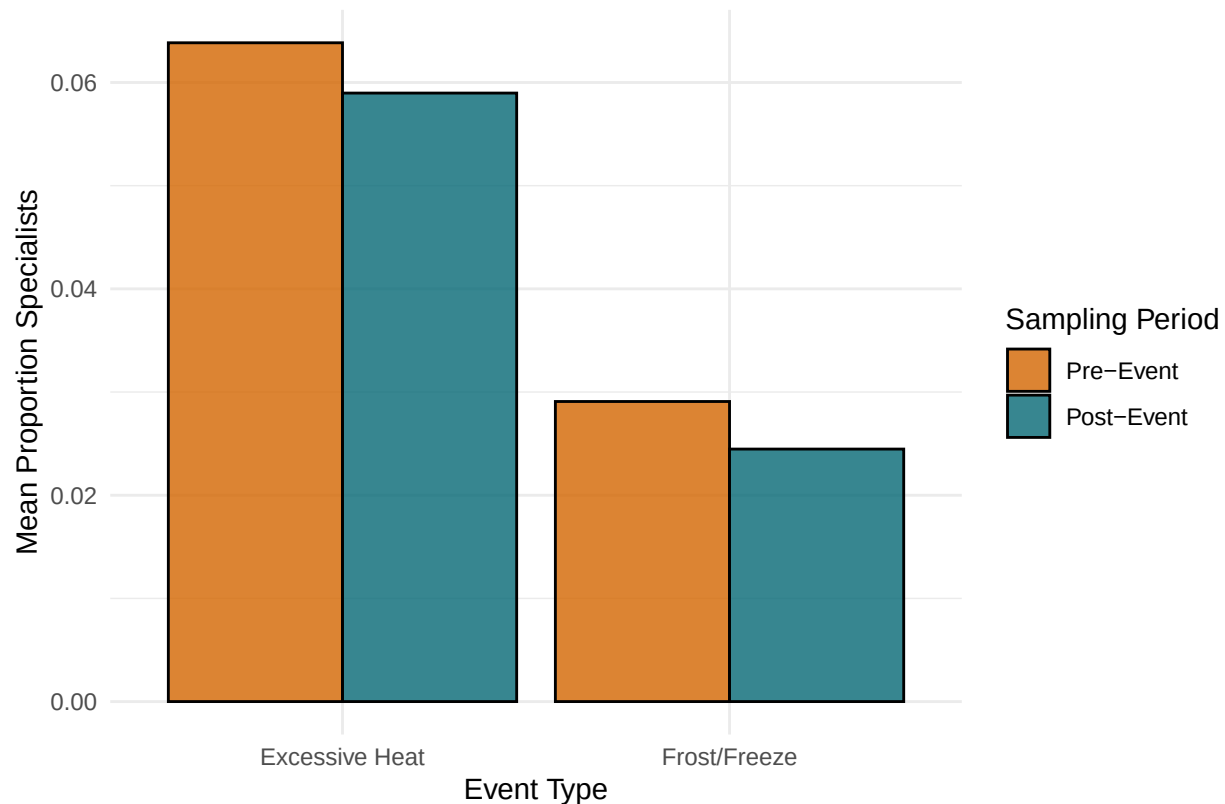
structure_plot_data$period <- factor(structure_plot_data$period, levels = c("mean_pre_prop_spec", "mean_post_prop_spec"))

structure_plot_data_filtered <- structure_plot_data %>%
  filter(EVENT_TYPE != "Winter Storm")

ggplot(structure_plot_data_filtered, aes(x = EVENT_TYPE, y = prop_spec, fill = period)) +
  geom_col(position = "dodge", color = "black", alpha = 0.8) +
  theme_minimal() +
  ggtitle("Bee Community Structure (Specialists vs Generalists) Before and After Extreme Weather Events")
```

```
xlab("Event Type") +
ylab("Mean Proportion Specialists") +
labs(fill = "Sampling Period") +
theme(plot.title = element_text(size = 10, face = "bold", hjust = 0.5)) +
scale_fill_manual(values=c("#D46600", "#056875"))
```

Community Structure (Specialists vs Generalists) Before and After Extreme Weather Events



```
# Calculate Percentage Decrease in Specialist Proportion After Heat
((structure_plot_data_filtered$prop_spec[1] - structure_plot_data_filtered$prop_spec[2]) / structure_pl
```

```
## [1] 7.632507
```

```
# Calculate Percentage Decrease in Specialist Proportion After Freezing Temperatures
((structure_plot_data_filtered$prop_spec[3] - structure_plot_data_filtered$prop_spec[4]) / structure_pl
```

```
## [1] 15.85499
```

Limitations / Potential Problems

Our storm data was limited to around 600 instances, and some counties were not represented at all, such as Baker, Umatilla, and Coos. When we analyze diversity among bees after storms, the lack of diversity data in specific locations may impact diversity metrics in certain bee communities. Additionally, our storm data is not all-encompassing. There were only 600 instances of combined freezing temperatures and heat

(over 100 degrees F). This is obviously not all instances of extreme temperatures recorded, so our data is not taking into account all instances of extreme temperature. The discrepancy found between counties with bee records and storms is obvious if you look at the ArcGIS Pro layout found in the repo. Additionally, because the NOAA data does not contain specific locations (e.g. xy coordinates) we were not able to compute where exactly temperature events occurred. For example, county regions are very large, and climate data from weather stations may vary widely between the northern tip of a country and the southern tip. This could lead to bees being recorded as “storm bees” when they were actually not storm bees since they were in a separate part of the county.

After speaking with Dr. Lincoln Bartlett, one of the researchers in charge of maintaining the Oregon Bee Atlas data, we learned that the Shannon Diversity index is not typically very strong at calculating diversities over such a large dataset. With so many bees in the data, the diversity is not great because the values will almost always be very large (between 4 and 5) and therefore it will be harder to find diversities that small in subsets of the data, like the storms. This is most likely why our P values were all incredibly small ($P = 0$).

Also, bees counts were collected by hand in the field, so it is natural that extreme weather events would impact the number of people collecting samples in the field. This could greatly affect sample counts of bees and bias the diversity data. It is possible that bees are not as impacted by extreme weather events as our results indicate due to the decreased sampled efforts following extreme weather events.

There is also limitations on the extent of this impact. When comparing before and after events, we focused on samples 30 days pre and post events. This results show short term changes in diversity and community composition and do not fully reflect long term trends which may be different from the immediate impacts of these events.

Conclusion

Our analysis reveals evidence that extreme weather events can alter bee community structure in ways that reduce biodiversity and favor generalist taxa. Shannon diversity values declined in the 30 days following extreme temperature events, suggesting that these disturbances may disrupt foraging behavior and shift communities toward genera with broader dietary flexibility. This could indicate that specialists are becoming more stressed and susceptible to impacts.

These results, however, should be looked at with caution. Although we concluded that, with $P = 0$ confidence, bee communities are less diverse according to the Shannon diversity index after extreme temperature events, the strength of our conclusions is limited by both the resolution of the temperature data and uncertainties surrounding the hand-collected data. We can't be certain of any conclusions as floral availability, weather station proximity, and collector effort may also have influenced the recorded community structure. More analysis, with higher resolution temperature data and standardized sampling methods would be needed to draw future conclusions. Even so, this analysis highlights the importance of examining short-term climate extremes rather than focusing only on long-term warming trends.

Overall, our findings show that continued monitoring of bee communities after extreme weather events is important and needed. Detecting shifts in community structure may help identify at-risk species and support future restoration efforts and strategies aimed at strengthening bee community resilience.

References

- Danforth, B. N., Minckley, R. L., Neff, J. L., & Fawcett, F. (2019). *The solitary bees: Biology, evolution, conservation*. Princeton University Press.
- Fowler, J. (2016). Specialist Bees of the Northeast: Host Plants and Habitat Conservation. *Northeastern Naturalist*, 23(2), 305–320. <https://doi.org/10.1656/045.023.0210>

- Franzén, M., & Öckinger, E. (2012). Climate-driven changes in pollinator communities: The return of the dark side of cold winters. *Journal of Insect Conservation*, 16(2), 215–222. <https://doi.org/10.1007/s10841-011-9410-y>
- Goulson, D. (2010). *Bumblebees: Behaviour, ecology, and conservation*. Oxford University Press.
- Goulson, D., Nicholls, E., Botías, C., & Rotheray, E. L. (2015). Bee declines driven by combined stress from parasites, pesticides, and lack of flowers. *Science*, 347(6229), 1255957. <https://doi.org/10.1126/science.1255957>
- Heinrich, B. (1979). *Bumblebee economics*. Harvard University Press.
- Intergovernmental Panel on Climate Change. (IPCC) (2021). *Climate change 2021: The physical science basis* (V. Masson-Delmotte, P. Zhai, A. Pirani, et al., Eds.). Cambridge University Press. <https://doi.org/10.1017/9781009157896>
- Michener, C. D. (2000). *The bees of the world*. Johns Hopkins University Press.
- Mishra, M., Bhunia, P., Shubham, S., Sen, R., Bhunia, R., Gulia, J., Mondal, T., Zidane, Z., Saha, T., Chacko, A., Raj, R., Vyshak, S. S., & Kaushal, L. (2023). The impact of weather change on honey bee populations and disease. *Journal of Advanced Zoology*, 44(S7), 180–190. <https://doi.org/10.17762/jaz.v44iS7.2755>
- Oregon Climate Change Research Institute. (2023). *The Fifth Oregon Climate Assessment*. Oregon State University, Corvallis, OR. <https://www.climatechange.oregonstate.edu/oregon-climate-assessments>
- Pereboom, J. J., & Biesmeijer, J. C. (2003). Thermal constraints for stingless bee foragers: The importance of body size and coloration. *Oecologia*, 137(1), 42–50. <https://doi.org/10.1007/s00442-003-1333-6>
- Potts, S. G., Imperatriz-Fonseca, V., Nguyen, H. T., Bartuska, A., Medellín, R. A., Baste, I. A., Oteng-Yeboah, A., Watson, R., et al. (Eds.). (2016). *The assessment report of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services on pollinators, pollination and food production*. Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services.
- Praz, C. J., Müller, A., & Dorn, S. (2018). Host-plant specialization in solitary bees: A case for polylecty evolution. *Biological Reviews*, 93(4), 1693–1716. <https://doi.org/10.1111/brv.12414>
- Russo, L. (2016). Positive and negative impacts of non-native bees on native ecosystems. *Annual Review of Entomology*, 61, 13–31. <https://doi.org/10.1146/annurev-ento-010715-023719>
- Thornton, P. K., Whitfield, S., & Challinor, A. J. (2023). Climate extremes and their implications for global food systems. *Annual Review of Environment and Resources*, 48, 203–229. <https://doi.org/10.1146/annurev-environ-120120-021013>
- Wilson, J. S., & Carril, O. M. (2015). *The bees in your backyard: A guide to North America's bees*. Princeton University Press.
- Woods, H. A., Dillon, M. E., & Pincebourde, S. (2019). The roles of microclimatic diversity and of behavior in mediating the responses of ectotherms to climate change. *Journal of Thermal Biology*, 84, 80–92. <https://doi.org/10.1016/j.jtherbio.2019.06.003>
- Danforth, Bryan N., Robert L. Minckley, Jesse L. Neff, and Francis Fawcett. 2019. *The Solitary Bees: Biology, Evolution, Conservation*. Princeton University Press.
- Fowler, James. 2016. “Specialist Bees of the Northeast: Host Plants and Habitat Conservation.” *Northeastern Naturalist* 23 (2): 305–20. <https://doi.org/10.1656/045.023.0210>.
- Franzén, Markus, and Erik Öckinger. 2012. “Climate-Driven Changes in Pollinator Communities: The Return of the Dark Side of Cold Winters.” *Journal of Insect Conservation* 16 (2): 215–22. <https://doi.org/10.1007/s10841-011-9410-y>.
- Goulson, Dave. 2010. *Bumblebees: Behaviour, Ecology, and Conservation*. Oxford University Press.
- Goulson, Dave, Elizabeth Nicholls, Cristina Botías, and Emma L. Rotheray. 2015. “Bee Declines Driven by Combined Stress from Parasites, Pesticides, and Lack of Flowers.” *Science* 347 (6229): 1255957. <https://doi.org/10.1126/science.1255957>.

- Heinrich, Bernd. 1979. *Bumblebee Economics*. Harvard University Press.
- Intergovernmental Panel on Climate Change (IPCC). 2021. *Climate Change 2021: The Physical Science Basis*. Edited by V. Masson-Delmotte, P. Zhai, A. Pirani, et al. Cambridge University Press. <https://doi.org/10.1017/9781009157896>.
- Michener, Charles D. 2000. *The Bees of the World*. Johns Hopkins University Press.
- Mishra, Manas, Prasanta Bhunia, S. Shubham, R. Sen, R. Bhunia, J. Gulia, T. Mondal, et al. 2023. “The Impact of Weather Change on Honey Bee Populations and Disease.” *Journal of Advanced Zoology* 44 (S7): 180–90. <https://doi.org/10.17762/jaz.v44iS7.2755>.
- Oregon Climate Change Research Institute. 2023. “The Fifth Oregon Climate Assessment.” Corvallis, OR: Oregon State University. <https://www.climatechange.oregonstate.edu/oregon-climate-assessments>.
- Pereboom, Jan J., and Jacobus C. Biesmeijer. 2003. “Thermal Constraints for Stingless Bee Foragers: The Importance of Body Size and Coloration.” *Oecologia* 137 (1): 42–50. <https://doi.org/10.1007/s00442-003-1333-6>.
- Potts, Simon G., Vera Imperatriz-Fonseca, Hong Thi Nguyen, Alton Bartuska, Rodrigo A. Medellín, Ivar A. Baste, Alfred Oteng-Yeboah, Robert Watson, et al., eds. 2016. *The Assessment Report of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services on Pollinators, Pollination and Food Production*. Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity; Ecosystem Services.
- Praz, Christophe J., Andreas Müller, and Silvia Dorn. 2018. “Host-Plant Specialization in Solitary Bees: A Case for Polylecty Evolution.” *Biological Reviews* 93 (4): 1693–1716. <https://doi.org/10.1111/brv.12414>.
- Thornton, Philip K., Sara Whitfield, and Andrew J. Challinor. 2023. “Climate Extremes and Their Implications for Global Food Systems.” *Annual Review of Environment and Resources* 48: 203–29. <https://doi.org/10.1146/annurev-environ-120120-021013>.
- Wilson, Joseph S., and Olivia M. Carril. 2015. *The Bees in Your Backyard: A Guide to North America’s Bees*. Princeton University Press.
- Woods, H. Arthur, Michael E. Dillon, and Sylvain Pincebourde. 2019. “The Roles of Microclimatic Diversity and of Behavior in Mediating the Responses of Ectotherms to Climate Change.” *Journal of Thermal Biology* 84: 80–92. <https://doi.org/10.1016/j.jtherbio.2019.06.003>.